Basics of Data Science

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Homework 0

Topics: Supervised Learning Foundations, Bias-Variance Decomposition, Augmented Ridge Regression, Softmax Concavity, and Cross-Validation Leakage

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Problem 1: Expected Prediction Error & Bias-Variance in k-NN [25 points]

Consider the classical nonparametric regression setting. Suppose we have a training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ where the scalar targets are generated by the model:

$$y = f(x) + \epsilon, \quad \text{with } \mathbb{E}[\epsilon \mid x] = 0 \text{ and } \operatorname{Var}(\epsilon \mid x) = \sigma^2,$$

where the additive noise ϵ is independent across observations and independent of the input vector $x \in \mathbb{R}^p$.

For a fixed query test point $x_0 \in \mathbb{R}^p$, consider estimating the unknown regression function $f(x_0)$ using the 1-Nearest-Neighbor (1-NN) estimator:

$$\hat{f}_1(x_0) = y_{(1)},$$

where $x_{(1)} \in \{x_1, \ldots, x_N\}$ denotes the training input closest to x_0 under Euclidean distance, and $y_{(1)}$ is its associated observed target.

(a) **[9 points]** Under the standard squared-error loss function $L(y, \hat{f}) = (y - \hat{f})^2$, show that the conditional Expected Prediction Error at x_0, evaluated with respect to a new independent target observation $y_0 = f(x_0) + \epsilon_0$ and conditioned on the training inputs $\{x_i\}_{i=1}^N$, decomposes exactly as:

$$\mathrm{EPE}(x_0) \equiv \mathbb{E}_{\mathcal{D},\epsilon_0}\left[(y_0 - \hat{f}_1(x_0))^2 \;\middle|\; x_0, \{x_i\}_{i=1}^N\right] = \sigma^2 + \left[f(x_0) - f(x_{(1)})\right]^2 + \sigma^2.$$

Explicitly identify each of the three terms in this decomposition (the irreducible error, the squared model bias, and the estimator sampling variance).

(b) **[9 points]** Generalize this result to the k-Nearest-Neighbor (k-NN) averaging estimator:

$$\hat{f}_k(x_0) = \frac{1}{k}\sum_{\ell=1}^{k} y_{(\ell)},$$

where $x_{(1)}, \ldots, x_{(k)}$ are the k closest training points to x_0. Derive the explicit formula for $\mathrm{EPE}(x_0; k)$ as a function of k, the underlying function values $\{f(x_{(\ell)})\}_{\ell=1}^k$, and the noise variance σ^2.

(c) **[7 points]** Analyze the behavior of $\mathrm{EPE}(x_0; k)$ as k varies from 1 to N:

- Explain mathematically what happens to the bias term and the variance term when $k = 1$ versus $k = N$.
- Relate this behavior to the fundamental concepts of model complexity, underfitting, and overfitting.
- Briefly discuss why the bias term fails to vanish in high dimensions ($p \gg 1$) even for $k = 1$ (the curse of dimensionality).

Problem 2: Ridge Regression as an Augmented OLS System [25 points]

In high-dimensional and collinear data regimes, Ordinary Least Squares can suffer from severe variance inflation or non-invertibility of $X^T X$. Ridge regression stabilizes parameter estimates by penalizing the L_2-norm of the regression coefficients:

$$\min_{\beta \in \mathbb{R}^p} \left\{ \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \right\},$$

where $\lambda > 0$ is a fixed regularization hyperparameter.

(a) **[5 points]** Define an augmented feature matrix $X^* \in \mathbb{R}^{(N+p)\times p}$ and an augmented response vector $y^* \in \mathbb{R}^{N+p}$ by vertically stacking:

$$X^* = \begin{bmatrix} X \\ \sqrt{\lambda} I_p \end{bmatrix}, \quad y^* = \begin{bmatrix} y \\ 0_p \end{bmatrix},$$

where I_p is the $p \times p$ identity matrix and 0_p is a p-dimensional vector of zeros.

(b) **[6 points]** Consider setting up an ordinary (unregularized) least squares problem on this augmented data:

$$\min_{\beta \in \mathbb{R}^p} \|y^* - X^* \beta\|_2^2.$$

Show algebraically that this augmented OLS objective is identically equal to the Ridge regression objective function.

(c) **[7 points]** Show that the closed-form OLS solution on the augmented dataset:

$$\hat{\beta}^* = (X^{*T} X^*)^{-1} X^{*T} y^*$$

reproduces the regularized Ridge regression estimator:

$$\hat{\beta}_{\text{ridge}} = (X^T X + \lambda I_p)^{-1} X^T y.$$

(d) **[7 points]** Prove that for any $\lambda > 0$, the matrix $X^{*T} X^*$ is strictly positive definite:

$$v^T (X^{*T} X^*) v > 0 \quad \forall v \in \mathbb{R}^p \setminus \{0_p\},$$

regardless of whether $N < p$ or whether the original matrix X has linearly dependent columns. Explain how this algebraic property guarantees that the Ridge normal equations always possess a unique and numerically stable solution.

Problem 3: Multiclass Softmax Regression & Log-Likelihood Concavity[25 points]

Consider a classification problem with $K \geq 2$ mutually exclusive classes. For each sample $i \in \{1, \ldots, N\}$, the response $y_i \in \{1, \ldots, K\}$ is encoded via a 1-of-K indicator vector $y_i = [y_{i1}, \ldots, y_{iK}]^T \in \{0,1\}^K$ such that $\sum_{k=1}^K y_{ik} = 1$.

The multiclass logistic regression model (Softmax classifier) models class posterior probabilities as:

$$p_k(x_i; \theta) \equiv P(Y = k \mid x_i) = \frac{\exp(\beta_k^T x_i)}{\sum_{\ell=1}^K \exp(\beta_\ell^T x_i)}, \quad k = 1, \ldots, K,$$

where $x_i \in \mathbb{R}^{p+1}$ (including the constant bias feature 1) and $\theta = [\beta_1^T, \ldots, \beta_K^T]^T \in \mathbb{R}^{K(p+1)}$ gathers all parameter vectors.

(a) **[5 points]** Formulate the conditional log-likelihood function $\ell(\theta)$ for an independently sampled training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$:

$$\ell(\theta) = \sum_{i=1}^N \sum_{k=1}^K y_{ik} \log p_k(x_i; \theta).$$

(b) **[7 points]** Derive the gradient vector of the log-likelihood with respect to an arbitrary class weight vector β_k ($k \in \{1, \ldots, K\}$), showing that:

$$\nabla_{\beta_k} \ell(\theta) \equiv \frac{\partial \ell(\theta)}{\partial \beta_k} = \sum_{i=1}^N (y_{ik} - p_k(x_i; \theta))\, x_i.$$

Provide an intuitive physical interpretation of this gradient update rule in terms of prediction error and feature weighting.

(c) **[6 points]** Compute the block components of the Hessian matrix $\mathcal{H}_{k\ell} \equiv \frac{\partial^2 \ell(\theta)}{\partial \beta_k \partial \beta_\ell^T} \in \mathbb{R}^{(p+1)\times(p+1)}$, proving that:

$$\frac{\partial^2 \ell(\theta)}{\partial \beta_k \partial \beta_\ell^T} = -\sum_{i=1}^N p_k(x_i; \theta)\, (\delta_{k\ell} - p_\ell(x_i; \theta))\, x_i x_i^T,$$

where $\delta_{k\ell}$ is the Kronecker delta ($\delta_{k\ell} = 1$ if $k = \ell$, and 0 otherwise).

(d) **[7 points]** Prove that the negative Hessian $-\mathcal{H} = -\nabla^2 \ell(\theta)$ is positive semi-definite everywhere in parameter space:

$$u^T(-\mathcal{H})u \geq 0 \quad \forall u \in \mathbb{R}^{K(p+1)}.$$

Conclude that $\ell(\theta)$ is strictly concave (up to an invariant additive constant), guaranteeing that gradient ascent and Newton-Raphson methods converge to a unique global optimum.

Problem 4: Feature Selection & Data Leakage in Cross-Validation [25 points]

Data leakage during model validation is one of the most pervasive failure modes in machine learning and empirical data science.

Consider the following simulation experiment:

- A synthetic dataset contains $N = 50$ samples and $p = 5000$ predictors.
- All feature entries $X_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$ $(i = 1, \ldots, 50;\ j = 1, \ldots, 5000)$ are sampled completely independently of each other.
- The binary class labels $Y_i \in \{0, 1\}$ are assigned randomly by tossing an unbiased coin:

$$P(Y_i = 1) = P(Y_i = 0) = 0.5.$$

Because the response Y and all predictors X are completely independent, the true population Bayes optimal classifier achieves a generalization error rate of exactly $\text{Err}_{\text{true}} = 50\%$.

(a) **[9 points] (The Flawed Protocol: Data Leakage)**
A practitioner executes the following validation procedure:

1. Across the entire dataset of $N = 50$ samples, compute the sample correlation $r_j = \widehat{\text{Corr}}(X_j, Y)$ for each of the $p = 5000$ features.
2. Select the subset $\mathcal{S} \subset \{1, \ldots, p\}$ containing the $m = 100$ predictors with the largest absolute sample correlation $|r_j|$.
3. Discard the other 4900 features and evaluate a 1-Nearest-Neighbor or Linear Discriminant classifier using standard 10-fold cross-validation restricted to the 100 features in $\mathcal{S}$.

Explain mathematically and conceptually why this procedure will report a 10-fold cross-validation error rate near 0%, dramatically misleading the researcher into believing they have discovered a near-perfect predictive model.

(b) **[7 points] (Extreme Value Statistics of Spurious Correlation)**
When $N = 50$, what is the distribution of the sample correlation r_j under the null hypothesis of independence?
Given that $p = 5000 \gg N$, analyze the expected behavior of the extreme order statistic:

$$\mathbb{E}\left[\max_{1 \le j \le 5000} |r_j|\right].$$

Show why having a massive search space of features guarantees that several dozen features will exhibit strong empirical correlation with Y purely by random chance.

(c) **[9 points] (The Correct Protocol)**
Detail the precise, algorithmic protocol for conducting k-fold cross-validation correctly when supervised feature screening is required. Explicitly describe how the 100 features must be selected within each fold, and explain why this eliminates all information leakage between training and testing splits.